

The dataset: CIC-IoT-2023

Canadian Institute for Cybersecurity University of New Brunswick (2023)

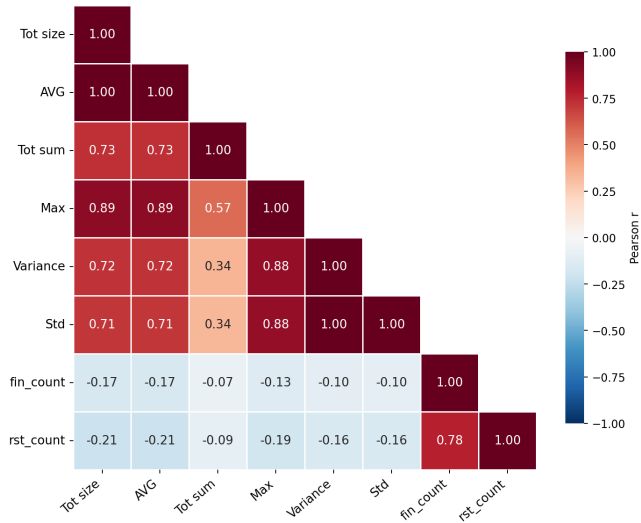
- ▶ 46.7M labelled flows · 105 real IoT devices
- ▶ 33 attacks across 7 families · smart-home testbed
- ▶ 39-feature public CSV — packet-header / flow metadata

Processing the 46M-row dataset

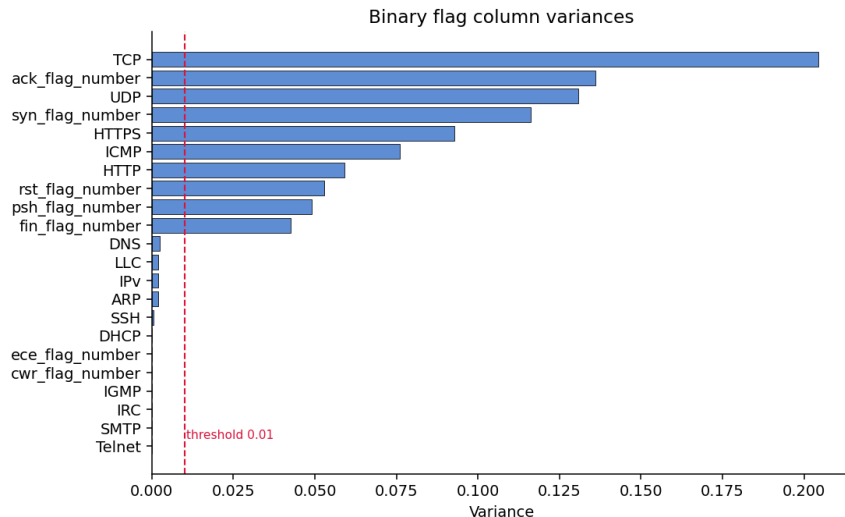
Polars + Apache Parquet

- ▶ **Polars** — lazy, multi-threaded columnar dataframes
- ▶ **Apache Parquet** — Zstd-compressed columnar storage

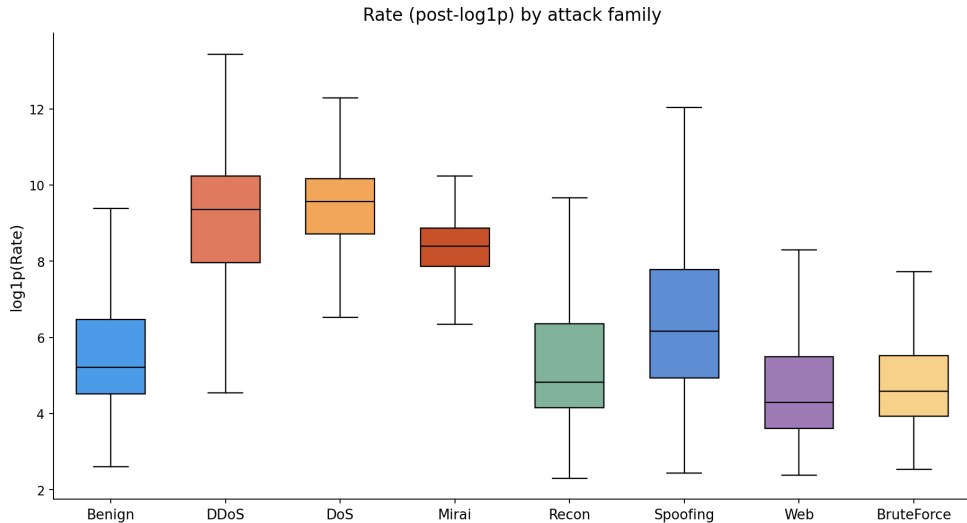
Feature correlation



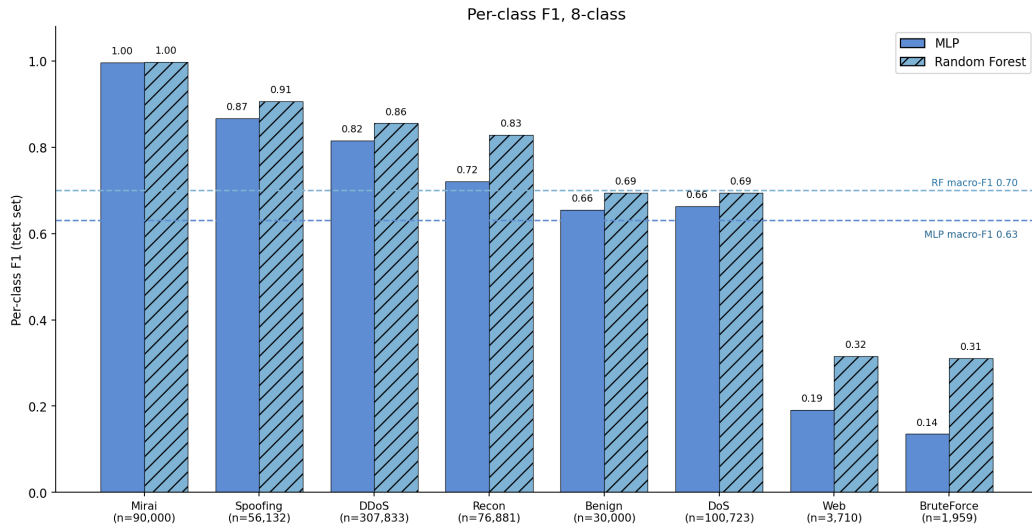
Feature variance



Feature separability — Rate by family



Application-layer attacks collapse



Comparable results only

